MILITARY STANDARD

PROTECTIVE FINISHING FOR MISSILE WEAPON SYSTEMS

TO ALL HOLDERS OF MIL-STD-186D:

1. THE FOLLOWING PAGES OF MIL-STD-186D HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
iii	10 November 1975	(REPRINTED WITHOUT CHANGE)	
iv	20 June 1980	(NEW PAGE)	
1	3 December 1976	(REPRINTED WITHOUT CHANGE)	
2	20 June 1980	2	3 December 1976
3	20 June 1980	3	3 December 1976
4	10 November 1975	(REPRINTED WITHOUT CHANGE)	
5	20 June 1980	5	3 December 1976
6	10 November 1975	(REPRINTED WITHOUT CHANGE)	
7	20 June 1980	7	3 December 1976
8	3 December 1976	(REPRINTED WITHOUT CHANGE)	
9	20 June 1980	9	3 December 1976
9a	20 June 1980	(NEW PAGE)	
10	10 November 1975	(REPRINTED WITHOUT CHANGE)	
12a	20 June 1980	12a	3 December 1976
12b	20 June 1980	12b	3 December 1976
12c	20 June 1980	12c	3 December 1976
12d	20 June 1980	12d	3 December 1976
13	10 November 1975	(REPRINTED WITHOUT CHANGE)	
14	20 June 1980	14	10 November 1975
15	20 June 1980	15	3 December 1976
16	10 November 1975	(REPRINTED WITHOUT CHANGE)	
23	10 November 1975	(REPRINTED WITHOUT CHANGE)	
24	20 June 1980	24	3 December 1976
29	20 June 1980	29	3 December 1976
30	3 December 1976	(REPRINTED WITHOUT CHANGE)	
33	20 June 1980	33	10 November 1975
34	20 June 1980	34	10 November 1975
41	20 June 1980	41	10 November 1975
42	20 June 1980	42	3 December 1976
43	10 November 1975	(REPRINTED WITHOUT CHANGE)	
44	20 June 1980	44	3 December 1976

MFFP

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:24Z
Check the source to verify that this is the current version before use.

MIL-STD-186D(MI)
NOTICE 2
20 June 1980

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
49	20 June 1980	49	10 November 1975
50	10 November 1975	(REPRINTED WITHOUT CHANGE)	
51	20 June 1980	51	10 November 1975
52	10 November 1975	(REPRINTED WITHOUT CHANGE)	

2. MAKE THE FOLLOWING PEN AND INK CHANGE:

Page ii, paragraph 2, delete "AMSMI-RCS" and substitute "DRSMI-ED".

3. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

4. Holders of MIL-STD-186D will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Custodian:
Army-MI

Review activities:
Army-ME, MR

Preparing activity:
Army-MI

Project No. MFFP-A208

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:24Z
Check the source to verify that this is the current version before use.

FOREWORD

The purpose of this standard is to establish minimum requirements for the protective finishing systems of rockets, guided missiles, all support equipment, and related materials.

For convenience in referencing, all procedures, whether they merely clean a surface, deposit a film, or perform some other desirable function, are regarded as "Finishes".

Examples of use of this document are as follow: Assume a part is to be finished with chromated cadmium plate, 0.0003 inch thick. Turning to table III, Metallic Coatings, find the designation for this finish to be code 305. The instructions on the drawing would be: "Code 305 MIL-STD-186." It is unnecessary to mention any preliminary steps such as cleaning, because specification QQ-P-416, Plating, Cadmium (Electrodeposited), referenced in code 305, provides for this.

Assume that an aluminum item is to be finished with semigloss enamel. According to table IV, Organic Finishes, the appropriate code might be code 425. In order to designate the color, the drawing would reference code 425, black 27038, or whichever color might be desired. Assume the preparation for painting to be wash primer, code 401, and the primer to be code 402; therefore, the complete process would require cleaning, code 106 from table I, Cleaning Methods, with the drawing instructions reading: "Finish codes 106/401/402/425, black 27038, in accordance with MIL-STD-186."

If circumstances require that parts of an assembly be primed separately and be given final coats after assembly, the instruction might read:

> "Finish Codes: 106/401/402/425, MIL-STD-186, prime before assembly."

Compliance with this standard will promote uniformity in the protective finishing of rockets, guided missiles, all support equipment, and related materials. Compliance with the standard does not relieve the supplier of his responsibility to assure satisfactory operation after storage as specified in his contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:24Z
Check the source to verify that this is the current version before use.

CONTENTS

			Page
Paragraph	1.	SCOPE	1
	1.1	Scope	1
	1.2	Selection of finishing systems	1
	1.2.1	Cross reference	1
	2.	APPLICABLE DOCUMENTS	1
	3.	DEFINITIONS	9
	4.	GENERAL REQUIREMENTS	10
	5.	DETAIL REQUIREMENTS	12d
	6.	INSPECTION	27a

TABLES

Table	I	Cleaning method	31
	II	Surface treatments	32
	III	Metallic coatings	35
	IV	Organic coatings	42
	V	Sealing and bonding	47
	VI	Encapsulants and potting compounds	49
	VII	Lubrication and preservation	50
	VIII	Miscellaneous	51
	IX	Finish cross reference "B" issue	52
	X	Galvanic couples	54
	XI	Group amplification of galvanic couples	55

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:24Z
Check the source to verify that this is the current version before use.